EIA-930 Data Pipeline Documentation

Comprehensive M-Code ETL Reference for Hourly Grid Balance, Generation Sources, Interchange & Sub-Regions

4 Pipelines	Q1 Focus	EIA-930
Power Query M Transformation Workflows	Time Filtered (Jan – Mar 2026)	US Energy Information Administration Grid Dataset

1. EXECUTIVE SUMMARY & ARCHITECTURE OVERVIEW

This document provides an exhaustive technical breakdown of the Power Query (M Language) ETL pipelines constructed for processing raw EIA-930 hourly electric power grid balance data. The dataset spans hourly balancing authority metrics across the United States. The transformation logic cleans raw CSV ingest, eliminates imputed and unadjusted data redundancy, standardizes date-time types, performs conditional demand calculations, unpivots fuel sources, and filters temporal scope to the target analysis period (Q1: January to March).

2. QUERY 1: MAIN BALANCE QUERY (DEMAND & GENERATION SUMMARY)

The `Main Balance Query` extracts macro-level power grid operational metrics including Demand Forecast, Adjusted Net Generation, Adjusted Total Interchange, and computes a fallback Demand values.

Step Name	Transformation Function	Technical Purpose & Logic
`Source`	`Csv.Document(File.Contents(...))`	Ingests raw CSV file containing 65 initial columns using Windows-1252 encoding.
`Promoted Headers`	`Table.PromoteHeaders`	Promotes row 1 strings into scalar table column header names.
`Changed Type`	`Table.TransformColumnTypes`	Casts columns into primitive types (text, date, datetime, Int64).
`Reordered Columns`	`Table.ReorderColumns`	Relocates `Region` alongside `Balancing Authority` for logical grouping.
`Removed Columns (raw & imputed)`	`Table.RemoveColumns`	Prunes 35 redundant raw fuel generation columns & imputed columns.
`Removed Columns has 0 and blanks`	`Table.RemoveColumns`	Removes unadjusted raw fields (`Demand` , `Net Gen` , `Interchange`) & empty fuels.
`Remove last 3 monthes (4,5,6)`	`Table.SelectRows`	Filters out months 4, 5, and 6 using `Date.Month(Date.From([Data Date]))` .

Key Objective of Query 1:

Isolate cleaned hourly demand and net generation metrics at the Balancing Authority level while removing raw uncalibrated readings and automated EIA imputations.

2. QUERY 1: MAIN BALANCE QUERY (CONTINUED)

Continued detailed step-by-step breakdown of transformations for the primary Balance query:

Step Name	Transformation Function	Technical Purpose & Logic
`Changed Type1 - local time`	`Table.TransformColumnTypes`	Converts local time column to `type time` and fuel source metrics to `type number`.
`Replaced Value - Empty cells...`	`Table.ReplaceValue`	Replaces `null` values across all 14 fuel generation columns with `0`.
`Removed Columns Sum(Valid)`	`Table.RemoveColumns`	Drops `Sum(Valid DIBAs) (MW)` column from dataset.
`Removed Other Columns - power sources`	`Table.SelectColumns`	Keeps only core operational attributes: Authority, Region, Date, Hour, Forecast, Generation, Interchange, Demand.
`Changed Type1`	`Table.TransformColumnTypes`	Enforces `type number` on demand forecast, net generation, interchange, and adjusted demand.
`Added Custom Demand`	`Table.AddColumn`	Evaluates custom logic to populate missing demand values dynamically.
`Renamed Columns`	`Table.RenameColumns`	Renames `Custom` column to `Demand(MW)(Adjusted)`.
`Removed Columns Demand & Changed Type2`	`Table.RemoveColumns` / `TransformColumnTypes`	Removes original adjusted demand column and sets final column datatypes.

CUSTOM CONDITIONAL LOGIC FOR DEMAND IMPUTATION

To ensure 100% completeness of hourly demand data without relying on synthetic imputation, a custom M expression was executed. When `Demand (MW) (Adjusted)` is missing, it is derived using grid physics: **Demand = Net Generation - Total Interchange**.

```
    each if [#"Demand (MW) (Adjusted)"] <> null then [#"Demand (MW) (Adjusted)"] else if [#"Net Generation (MW) (Adjusted)"] <> null and [#"Total Interchange (MW) (Adjusted)"] <> null and [#"Net Generation (MW) (Adjusted)"] - [#"Total Interchange (MW) (Adjusted)"] >= 0 then [#"Net Generation (MW) (Adjusted)"] - [#"Total Interchange (MW) (Adjusted)"] else null
```

Data Quality Guardrail:

The check `[Net Generation] - [Total Interchange] >= 0` prevents physical anomalies (e.g., negative demand values resulting from bad reporting sensor spikes during interchange outages).

3. QUERY 2: GENERATION SOURCES UNPIVOTED QUERY

The `Generation Sources Unpivoted Query` converts fuel source generation columns into a normalized relational format suitable for BI dimensional modeling.

Step Name	Transformation Function	Technical Purpose & Logic
`Source` through `Replaced Value`	Shared Initial Steps	Executes data loading, type conversion, Q1 filtering, and null-to-zero replacement across fuel columns.
`Unpivoted Columns - Sources' Net generation`	`Table.UnpivotOtherColumns`	Unpivots all 14 adjusted generation source columns into attribute-value pairs while anchoring dimensional keys.
`Renamed Columns`	`Table.RenameColumns`	Renames the default `Attribute` column to `Net Generation (MW)`.
`Replaced Value1`	`Table.ReplaceValue`	Strips text prefix `"Net Generation (MW) "` from fuel attribute labels.
`Replaced Value`	`Table.ReplaceValue`	Strips text suffix `"(Adjusted)"` to yield clean fuel category names (e.g., *Coal*, *Natural Gas*, *Solar*).
`Changed Type1`	`Table.TransformColumnTypes`	Ensures numeric datatypes across metric columns.
`Added Custom`	`Table.AddColumn`	Re-applies conditional demand derivation logic.
`Reordered Columns1`	`Table.ReorderColumns`	Organizes primary key attributes followed by fuel source type and generation megawatt values.
`Removed Columns`	`Table.RemoveColumns`	Removes redundant balance totals, retaining only unpivoted fuel source rows.

BENEFITS OF FUEL UNPIVOTING

> Unpivoting 14 wide fuel columns into a single attribute-value pair reduces schema rigidity. Instead of 14 separate measure formulas in Power BI/DAX, analysts can construct a single dynamic measure: `SUM('Generation Sources'[Value])` filtered by fuel category slicing.

4. QUERY 3: INTERCHANGE DATA QUERY (INTER-BA TRANSFERS)

Processes power transfers between balancing authorities and directly interconnected balancing authorities (DIBAs).

Step Name	Transformation Function	Technical Purpose & Logic
`Source`	`Csv.Document(File.Contents(...))`	Ingests raw CSV file `EIA930_INTERCHANGE_2026_Jan_Jun.csv` (9 columns).
`Promoted Headers & Changed Type`	`PromoteHeaders` / `TransformColumnTypes`	Promotes headers and sets column types for BA, DIBA, Interchange (MW), Date, and Regions.
`Remove last 3 monthes(4,5,6)`	`Table.SelectRows`	Filters out months 4, 5, and 6 to maintain quarter alignment with the main balance queries.
`Changed Type1`	`Table.TransformColumnTypes`	Converts `Local Time at End of Hour` column specifically to `type time`.

5. QUERY 4: SUB-REGION DATA QUERY (REGIONAL DEMAND GRANULARITY)

Tracks local electricity demand broken down by sub-regional zones within larger balancing authority territories.

Step Name	Transformation Function	Technical Purpose & Logic
`Source`	`Csv.Document(File.Contents(...))`	Ingests raw CSV file `EIA930_SUBREGION_2026_Jan_Jun.csv` (7 columns).
`Promoted Headers & Changed Type`	`PromoteHeaders` / `TransformColumnTypes`	Promotes headers and assigns column types for BA, Sub-Region, Data Date, and Demand (MW).
`Remove last 3 monthes(4,5,6)`	`Table.SelectRows`	Filters out months 4, 5, and 6 to keep dataset aligned to Q1.
`Changed Type1`	`Table.TransformColumnTypes`	Adjusts time formatting on local hour fields to `type time`.

6. DATA MODEL STAR-SCHEMA RELATIONSHIP ARCHITECTURE

Recommended Primary Relationships:

- **Composite Join Key:** `Balancing Authority` + `Data Date` + `Hour Number` links the Main Balance Query to Interchange & Sub-Region tables.
- **Date Dimension:** Connect `Data Date` across all queries to a centralized `DateTable` for standardized time-intelligence DAX functions.